

GRAIN AND TILLER: LOSS-INFORMED RATIONAL TRANSFORMER UPDATES

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ABSTRACT

1 INTRODUCTION

2 RELATED WORK

3 METHOD

We introduce two coupled components. First, GRAIN learns a normalized rational response in each feed-forward feature group. Second, response statistics interpolate between a Muon direction and an orthogonal, factorized adaptive direction of the same norm. Finally, TILLER measures the loss derivative of every paired group direction, estimates their cross-coordinate interactions, and selects signed coefficients under a fixed budget for the GRAIN structured directions.

Let a Transformer have L feed-forward blocks, G rational groups per block, model width d , and expanded rational width $H_r = Gw$. We index the $C = LG$ layer-group coordinates by $i = (\ell, g)$. Matrix inner products and norms are Frobenius inner products and norms.

3.1 GRAIN: A NORMALIZED GROUPWISE RATIONAL ACTIVATION

Gated feed-forward networks use multiple projections to create a multiplicative feature map (Dauphin et al., 2017; Shazeer, 2020). GRAIN applies a learned scalar rational function shared within each feature group. For an input $x \in \mathbb{R}^d$, one block is

$$z = Ax, \quad h = \Phi(z), \quad y = Bh, \quad (1)$$

where $A \in \mathbb{R}^{H_r \times d}$ and $B \in \mathbb{R}^{d \times H_r}$. Partition z into G contiguous groups $z_g \in \mathbb{R}^w$. At layer ℓ , write $A_{\ell g} \in \mathbb{R}^{w \times d}$ for the corresponding row block of A_ℓ and $B_{\ell g} \in \mathbb{R}^{d \times w}$ for the corresponding column block of B_ℓ . Then

$$z_{\ell g} = A_{\ell g} x_\ell, \quad y_\ell = \sum_{g=1}^G B_{\ell g} \Phi_g(z_{\ell g}). \quad (2)$$

For each group, define

$$\rho_g = \sqrt{w^{-1} \|z_g\|_2^2 + \epsilon_{\text{act}}}, \quad u_g = z_g / \rho_g, \quad \epsilon_{\text{act}} = 10^{-6}. \quad (3)$$

Each group has a learned rational response with a degree-five numerator and degree-four denominator,

$$P_g(u) = \sum_{k=0}^5 a_{gk} u^k, \quad (4)$$

$$Q_g(u) = 1 + \sum_{k=1}^4 |b_{gk}| |u|^k, \quad (5)$$

$$\Phi_g(z_g) = \rho_g \phi_g(u_g), \quad \phi_g(u) = P_g(u) / Q_g(u), \quad (6)$$

where all functions and powers act elementwise. This combines learned rational activations (Molina et al., 2020; Boullé et al., 2020) with an internal RMS coordinate system (Zhang & Sennrich, 2019). Because $Q_g(u) \geq 1$, the response is finite for every real input.

For a SwiGLU intermediate width H , we set $H_r = 3H/2$. The two dense matrices in equation 1 then contain $2dH_r = 3dH$ weights, equal to the dense projection count of the three SwiGLU matrices. The activation adds $10G$ rational coefficients per block. This parameterization preserves the projection budget. The learned response supplies the Jacobian and feature statistics used below.

The normalized map has a diagonal-plus-rank-one Jacobian wherever the rational response is differentiable. Write $f_g = \phi_g(u_g)$, $d_g = \phi'_g(u_g)$, and $r_g = f_g - u_g \odot d_g$, all in $\mathbb{R}^w$. Direct differentiation gives

$$J_g \equiv \frac{\partial \Phi_g}{\partial z_g} = \text{Diag}(d_g) + \frac{1}{w} r_g u_g^\top, \quad J_g^\top v = d_g \odot v + u_g \text{ mean}(r_g \odot v). \quad (7)$$

Thus a group response and its pullback reduce to a diagonal operation and one rank-one correction. This structure lets us evaluate loss derivatives of matrix directions directly in activation space.

3.2 RESPONSE-AWARE STRUCTURED DIRECTIONS

Parent and adaptive directions. For each pair of group blocks $i = (\ell, g)$ and role $r \in \{A, B\}$, TILLER starts from a Nesterov–Muon direction p_i^r : with $\beta_1 = 0.95$, it forms

$$v_{t,i}^r = \beta_1 v_{t-1,i}^r + (1 - \beta_1) g_{t,i}^r, \quad m_{t,i}^r = \beta_1 v_{t,i}^r + (1 - \beta_1) g_{t,i}^r, \quad (8)$$

followed by a five-step Newton–Schulz polar map applied independently to each group block and Muon’s role-dependent RMS adjustment (Jordan et al., 2024; Bernstein & Newhouse, 2024; Liu et al., 2025). The groupwise map has nominal rank $G \min\{w, d\}$, whereas the corresponding full matrix has rank at most $\min\{H_r, d\}$. We therefore apply the calibration

$$\chi_{\text{grp}} = \sqrt{\frac{\min\{H_r, d\}}{G \min\{w, d\}}}, \quad (9)$$

which matches the squared Frobenius scale of the full-matrix polar direction. Writing $\mathcal{P}_5$ for this blockwise polar map, Muon’s adjustment is $\mu_{\text{mlp}} = 0.2 \sqrt{\max\{H_r, d\}}$ for either GRAIN matrix role. The parent group direction is $p_{t,i}^r = \mu_{\text{mlp}} \chi_{\text{grp}} \mathcal{P}_5(m_{t,i}^r)$. In parallel, TILLER maintains factorized row and column second moments, in the spirit of memory-efficient factored adaptation (Shazeer & Stern, 2018). For a grouped matrix with gradient g_i and Nesterov momentum m_i , suppressing role and time indices,

$$R_t = \beta_2 R_{t-1} + (1 - \beta_2) \text{rowsum}(g_i^2), \quad (10)$$

$$C_t = \beta_2 C_{t-1} + (1 - \beta_2) \text{colsum}(g_i^2), \quad (11)$$

$$V_{t,jk} = \frac{R_{t,j} C_{t,k}}{\max\{\sum_s R_{t,s}, \delta_{\text{fp}}\}} \frac{1}{1 - \beta_2^t}, \quad a_i = \mu_{\text{mlp}} \frac{m_i}{\sqrt{V_t} + \epsilon_{\text{opt}}}. \quad (12)$$

We set $\beta_2 = 0.95$ and $\epsilon_{\text{opt}} = 10^{-8}$. The same β_2 sets the time scale of the loss ledger in Section 3.4, aligning the adaptive-direction and response histories. The zero-total convention for the factorized estimate is given in Appendix F.

A response-derived mixing fraction. Two bounded summaries determine how much of the adaptive direction enters the update. At a response refresh, let $n \in \{1, \dots, K\}$ index the sampled positions and let $\xi_{n\ell} \in \mathbb{R}^d$ be the downstream cotangent at the output of block ℓ . Define

$$\text{pr}_A(J_{in}) = \frac{[\text{tr}(J_{in} J_{in}^\top)]^2}{w \|J_{in} J_{in}^\top\|_F^2}, \quad \Gamma_{in}^A = J_{in} J_{in}^\top, \quad (13)$$

$$\text{pr}_B(h_{in}) = \frac{\|h_{in}\|_2^4}{w \|h_{in}\|_4^4}, \quad \Gamma_{in}^B = h_{in} h_{in}^\top. \quad (14)$$

For the same normalized input, $\Gamma_{in}^{r,0}$ denotes the kernel obtained from the initial rational response. Let

$$\alpha_{in}^A = \|J_{in}^\top B_i^\top \xi_{n\ell}\|_2^2 / w, \quad \alpha_{in}^B = \|\xi_{n\ell}\|_2^2 / d \quad (15)$$

measure downstream loss sensitivity. Participation and congruence are

$$\pi_i^A = \frac{\sum_n \alpha_{in}^A \text{pr}_A(J_{in})}{\sum_n \alpha_{in}^A}, \quad \pi_i^B = \frac{\sum_n \alpha_{in}^B \text{pr}_B(h_{in})}{\sum_n \alpha_{in}^B}, \quad (16)$$

$$\kappa_i^r = \frac{\sum_n \alpha_{in}^r \langle \Gamma_{in}^r, \Gamma_{in}^{r,0} \rangle}{\sqrt{\left(\sum_n \alpha_{in}^r \|\Gamma_{in}^r\|_F^2 \right) \left(\sum_n \alpha_{in}^r \|\Gamma_{in}^{r,0}\|_F^2 \right)}}. \quad (17)$$

The two participation statistics answer role-specific versions of the same question: how broadly is the live response distributed? For the incoming role, pr_A is the normalized inverse participation ratio of the squared singular values of J_{in} ; it equals $1/w$ for a rank-one Jacobian and reaches one when all w singular values have equal magnitude. For the outgoing role, pr_B is the normalized inverse participation ratio of the coordinate energies $h_{in,j}^2$; it equals $1/w$ when all feature energy lies in one coordinate and reaches one when that energy is uniform. Congruence is the cosine between the live and initial kernels in a loss-weighted direct-sum space, so $1 - (\kappa_i^r)^2$ is the squared sine of their kernel angle. Their product therefore opens the adaptive tangent when the response is both broadly supported and different from its initialization. Zero-response conventions and bounds are given in Appendix B; whenever a comparison is undefined, we set its departure energy to zero. We allocate

$$e_i^r = \pi_i^r (1 - (\kappa_i^r)^2) \quad (18)$$

of the direction energy to an adaptive tangent.

A norm-preserving chord. Suppressing the role index, define

$$a_i^\perp = \begin{cases} a_i - \frac{\langle a_i, p_i \rangle}{\|p_i\|_F^2} p_i, & \|p_i\|_F > 0, \\ 0, & \|p_i\|_F = 0, \end{cases} \quad \sigma_i = \begin{cases} 1, & \langle g_i, a_i^\perp \rangle \geq 0, \\ -1, & \langle g_i, a_i^\perp \rangle < 0. \end{cases} \quad (19)$$

When $\|p_i\|_F > 0$ and $\|a_i^\perp\|_F > 0$, form the candidate

$$\tilde{q}_i = \sqrt{1 - e_i} p_i + \sigma_i \sqrt{e_i} \frac{\|p_i\|_F}{\|a_i^\perp\|_F} a_i^\perp. \quad (20)$$

The selected direction is

$$q_i = \begin{cases} \tilde{q}_i, & \|p_i\|_F \|a_i^\perp\|_F > 0, \langle g_i, p_i \rangle > 0, \langle g_i, \tilde{q}_i \rangle > 0, \text{ and the candidate is finite,} \\ p_i, & \text{otherwise.} \end{cases} \quad (21)$$

Because $a_i^\perp \perp p_i$, the two terms in equation 20 are orthogonal and $\|\tilde{q}_i\|_F^2 = (1 - e_i)\|p_i\|_F^2 + e_i\|p_i\|_F^2 = \|p_i\|_F^2$. Both selection branches therefore preserve the parent norm.

The construction has a geometric interpretation. A concentrated response supports little motion away from the polar direction, whereas a diffuse response supports a higher-dimensional tangent. Congruence distinguishes generic dimensionality from learned change. As congruence decreases, e_i^r assigns a larger norm fraction to the orthogonal adaptive component. The current gradient orients that component, permitting either sign.

3.3 LOSS-COORDINATE MEASUREMENTS

On optimizer step 1 and every eight steps thereafter, TILLER deterministically selects a probe set $\mathcal{P}_t$ of $K = 32$ example–token positions. The same positions index every coordinate at that refresh, so each row is a joint measurement across the network. For probe $k \in \mathcal{P}_t$ and coordinate $i = (\ell, g)$, let $x_{k,\ell}$ be the block input, $z_{k,i} = A_i x_{k,\ell}$, $h_{k,i} = \Phi_i(z_{k,i})$. Let $\xi_{k,\ell}$ be the corresponding output cotangent of the individual position loss, scaled by the clipping factor applied to the parameter gradient. Along the paired direction (q_i^A, q_i^B) ,

$$\delta z_{k,i} = q_i^A x_{k,\ell}, \quad \delta h_{k,i} = J_{k,i} \delta z_{k,i}, \quad \delta y_{k,i} = B_i \delta h_{k,i} + q_i^B h_{k,i}. \quad (22)$$

Pairing $\delta y_{k,i}$ with the downstream loss derivative gives

$$S_{t,k,i} = \langle q_i^A x_{k,\ell}, J_{k,i}^\top B_i^\top \xi_{k,\ell} \rangle + \langle q_i^B h_{k,i}, \xi_{k,\ell} \rangle. \quad (23)$$

At differentiability points, equation 23 is the clipped per-position directional contribution. Its row mean is a sampled estimate of the corresponding minibatch derivative, whereas the exact minibatch contraction is computed below from the parameter gradient. At zero coordinates, the score uses the derivative convention in Appendix A. The pullback in equation 7 evaluates its incoming term using groupwise contractions. Applying the same calculation to the decoupled decay direction and summing over coordinates gives a decay-action vector $a_t^{\text{dec}} \in \mathbb{R}^K$.

Between functional refreshes, the two matrix roles provide the current minibatch directional derivatives

$$s_{t,i} = \langle g_{t,i}^A, q_{t,i}^A \rangle + \langle g_{t,i}^B, q_{t,i}^B \rangle. \quad (24)$$

Let $\nu = \|S_{\text{ref}}\|_F / \sqrt{K}$ be the RMS row norm of the latest functional measurement. Set

$$\gamma_t = \begin{cases} \nu / \|s_t\|_2, & \|s_t\|_2 > 0, \\ 0, & \|s_t\|_2 = 0, \end{cases} \quad S_t = \mathbf{1}_K (\gamma_t s_t)^\top, \quad a_t^{\text{dec}} = \mathbf{1}_K \gamma_t d_t^{\text{dec}}, \quad (25)$$

where d_t^{dec} is the current minibatch derivative along the complete GRAIN decay direction. This construction exactly preserves the trace of the latest functional geometry because $\|S_t\|_F^2 = K \gamma_t^2 \|s_t\|_2^2 = \|S_{\text{ref}}\|_F^2$ whenever $s_t \neq 0$. Hence every step contributes the current first-order direction, while functional refreshes periodically restore probe-level cross-coordinate variation.

For coefficients $c \in \mathbb{R}^C$, the probe vector induced jointly by the structured direction and decoupled decay is $S_t c + a_t^{\text{dec}}$. Its squared response expands as

$$\frac{1}{K} \|S_t c + a_t^{\text{dec}}\|_2^2 = \frac{1}{K} \|S_t c\|_2^2 + \frac{2}{K} c^\top S_t^\top a_t^{\text{dec}} + \frac{1}{K} \|a_t^{\text{dec}}\|_2^2. \quad (26)$$

The last term is constant in c . The first two terms expose which coordinates repeatedly affect the same loss probes and whether their joint effect reinforces the fixed decay direction. They define a functional-response covariance surrogate rather than a Hessian model of the loss.

3.4 A ROBUST FREQUENT-DIRECTIONS LOSS LEDGER

The ledger tracks the two coefficient-dependent terms in equation 26. The score rows define a quadratic geometry over the $C = LG$ coordinates. TILLER represents this geometry by a low-rank factor and an isotropic remainder. It initializes $\mathcal{B}_1 = S_1$ and $\tau_1 = 0$. From the second step onward, it retains $\mathcal{B}_{t-1}$ with at most $r = 64$ rows and forms

$$\mathcal{A}_t = \begin{bmatrix} \sqrt{\beta_2} \mathcal{B}_{t-1} \\ \sqrt{1 - \beta_2} S_t \end{bmatrix}. \quad (27)$$

Following Frequent Directions and its robust isotropic completion (Ghashami et al., 2016; Luo et al., 2019), TILLER diagonalizes the row Gram matrix $\mathcal{A}_t \mathcal{A}_t^\top$. If Δ_t is the largest discarded eigenvalue, it shrinks every retained eigenvalue by Δ_t and adds $\Delta_t / (2K)$ to an isotropic tail. The persistent metric is

$$\hat{G}_t = \frac{1}{K} \mathcal{B}_t^\top \mathcal{B}_t + \tau_t I, \quad \tau_t = \beta_2 \tau_{t-1} + \frac{\Delta_t}{2K}. \quad (28)$$

The midpoint tail assigns half of the unrepresented eigenvalue interval to every direction. It therefore halves the worst-direction error of either endpoint completion, yielding the bound $\Delta_t / (2K)$ in Appendix D. The score-decay cross term is

$$r_t = \beta_2 r_{t-1} + (1 - \beta_2) S_t^\top a_t^{\text{dec}} / K. \quad (29)$$

The current transaction uses the uncompressed factor $\mathcal{A}_t$ and diagonal completion $\delta_t = \beta_2 \tau_{t-1} \mathbf{1}$; $(\mathcal{B}_t, \tau_t, r_t)$ is carried to the next step. At initialization, $\mathcal{A}_1 = S_1$, $\delta_1 = 0$, and $r_1 = S_1^\top a_1^{\text{dec}} / K$.

3.5 SIGNED EQUAL-BUDGET TRANSACTION

Let $c \in \mathbb{R}^C$ scale the paired directions and define $\omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2$. We work on the nondegenerate transaction domain $\omega_i > 0$ for every coordinate; this invariant is checked before whitening. The uncoordinated parent update is $c = \mathbf{1}$, whose squared norm is $\mathcal{E}_0 = \sum_i \omega_i$. Indeed, the parameter blocks associated with different coordinates are disjoint, so the structured update

$\Delta\theta(c) = -\eta\{c_i q_i^A, c_i q_i^B\}_{i=1}^C$ satisfies $\|\Delta\theta(c)\|_2^2 = \eta^2 \sum_i \omega_i c_i^2$. With current directional derivatives s_t , pre-compression factor $\mathcal{A}_t$, diagonal completion δ_t , score-decay cross term r_t , and scheduled step size η , TILLER forms the response penalty

$$\mathcal{R}_t(c) = \frac{1}{K} \|\mathcal{A}_t c\|_2^2 + \delta_t^\top (c \odot c) + 2r_t^\top c \quad (30)$$

and searches in a Krylov subspace for a projected positive-multiplier stationary candidate of the equal-budget model

$$\min_c F_t(c) = -\eta s_t^\top c + \frac{\eta^2}{2} \mathcal{R}_t(c), \quad (31)$$

$$\text{s.t.} \quad \sum_i \omega_i c_i^2 = \mathcal{E}_0. \quad (32)$$

The linear term rewards immediate descent. The quadratic term penalizes directions that the loss ledger predicts to interact strongly, including the effect of decoupled decay. The constraint gives the coordinated GRAIN direction the same aggregate squared norm as the parent before applying their common decay term. Because c is signed, a transaction may amplify, attenuate, or reverse individual coordinates.

After whitening $y = W^{1/2}c$, where $W = \text{Diag}(\omega)$, the stationarity equation is

$$(M_t + \lambda I)y = b_t, \quad M_t = W^{-1/2} (\mathcal{A}_t^\top \mathcal{A}_t / K + \text{Diag}(\delta_t)) W^{-1/2}, \quad b_t = W^{-1/2} (s_t / \eta - r_t). \quad (33)$$

For $b_t \neq 0$, define $\tilde{\mathcal{A}}_t = \mathcal{A}_t W^{-1/2}$ and $\tilde{\delta}_t = \delta_t \odot \omega$. A reorthogonalized Lanczos basis is constructed from products $v \mapsto \tilde{\delta}_t \odot v + \tilde{\mathcal{A}}_t^\top (\tilde{\mathcal{A}}_t v) / K$ (Lanczos, 1950). TILLER attempts the positive-multiplier regular branch, which reduces to a scalar secular equation for λ . The proposal is evaluated in the unprojected low-rank-plus-diagonal surrogate. It is retained when it satisfies the budget, lowers F_t relative to $c = 1$, and preserves positive layerwise gradient and momentum contractions

$$\sum_g c_{\ell g} \langle g_{\ell g}^r, q_{\ell g}^r \rangle > 0, \quad \sum_g c_{\ell g} \langle m_{\ell g}^r, q_{\ell g}^r \rangle > 0 \quad (r \in \{A, B\}). \quad (34)$$

The same contractions must also be positive for the parent coefficients $c = 1$. The parent coefficients are used when the implemented regular branch has no bracketed root or its proposal fails these checks. Appendix E derives this regular proposal.

3.6 COMPLETING THE TRANSFORMER UPDATE

For attention, the group statistics are summarized at each layer as

$$\pi_\ell^{\text{att}} = \sqrt{\left(G^{-1} \sum_g \pi_{\ell g}^A\right) \left(G^{-1} \sum_g \pi_{\ell g}^B\right)}, \quad \kappa_\ell^{\text{att}} = \sqrt{\kappa_\ell^A \kappa_\ell^B}, \quad (35)$$

where each κ_ℓ^r is formed by summing the numerator and two denominator terms of equation 17 over groups before normalization. With

$$e_\ell^{\text{att}} = \sqrt{\bar{\pi}_\ell^A \bar{\pi}_\ell^B} (1 - \kappa_\ell^A \kappa_\ell^B), \quad \bar{\pi}_\ell^r = G^{-1} \sum_g \pi_{\ell g}^r, \quad (36)$$

the participation factor requires broad support on both matrix roles, while the departure factor opens when either role has moved away from its initial response. These layerwise statistics gate the same norm-preserving chord for the QKV and output matrices. Attention heads are partitioned into blocks of four for the polar map; the stacked QKV direction is scaled by $1/\sqrt{3}$ to match its three constituent projections. The adaptive tangent is factorized at matrix level. Attention does not enter the LG coefficient transaction. Learned rational coefficients, embeddings, normalization parameters, and remaining tensors use AdamW (Loshchilov & Hutter, 2019).

The auxiliary state consists of factorized moments, response summaries, and a rank- r factor over the LG transaction coordinates. Its size is $O(LH_r + LGd + rLG)$. For fixed probe count, ledger rank, and Krylov dimension, both the ledger update and the transaction are linear in LG ; their dense linear algebra acts only on row or projected matrices of fixed dimension. The activation work is $O(LNH_r)$ for N activation positions, whereas the optimizer state does not depend on N . Appendix G gives the reduction identities for partitioned coordinates, and Appendix F states the complete update order.

4 EXPERIMENTS

5 DISCUSSION

6 CONCLUSION

AI USE STATEMENT

Generative AI tools assisted with code implementation, mathematical cross-checking, literature retrieval, and manuscript drafting. The authors reviewed the resulting material and take responsibility for the final content.

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A EXACT GRAIN CALCULUS

Proposition 1 (Pole freedom and Jacobian structure). *For every finite group input, $Q_g(u) \geq 1$. Whenever the absolute-value terms are differentiable, the Jacobian of equations 3 and 6 is equation 7. At $u_j = 0$, where the response need not be classically differentiable, we use the automatic-differentiation selection $\text{sign}(0) = 0$; throughout the paper, J_g denotes this selected derivative at such coordinates.*

Proof. Every term $|b_{gk}| |u|^k$ is nonnegative, hence $Q_g(u) \geq 1$. Suppress the group index and write $\rho = (w^{-1} z^\top z + \epsilon_{\text{act}})^{1/2}$ and $u = z/\rho$. Then

$$\nabla_z \rho = u/w, \quad \frac{\partial u}{\partial z} = \rho^{-1} (I - uu^\top/w). \quad (37)$$

Applying the product and chain rules to $\Phi(z) = \rho\phi(u)$ gives

$$J = \phi(u)u^\top/w + \text{Diag}(\phi'(u)) (I - uu^\top/w) \quad (38)$$

$$= \text{Diag}(d) + (\phi(u) - u \odot d) u^\top/w. \quad (39)$$

Transposition and multiplication by v yield the pullback in equation 7. $\square$

All groups begin from the same SiLU fit. In ascending powers, the coefficients are

$$(a_0^{\text{init}}, \dots, a_5^{\text{init}}) = (2.4915714620520875 \times 10^{-5}, 0.5000668663485823, 0.2499441908995771, \\ 0.0526200910151016, 0.005525765640115002, 0.00024199321178747251), \quad (40)$$

$$(b_1^{\text{init}}, \dots, b_4^{\text{init}}) = (0.00022554017910995938, 0.10511420350691994, \\ 2.8656992394109812 \times 10^{-5}, 0.0004816962110562025). \quad (41)$$

The ten coefficients subsequently learn independently in every group.

B RESPONSE GEOMETRY

At probe position n , the initialization kernels $\Gamma_{in}^{A,0}$ and $\Gamma_{in}^{B,0}$ evaluate the frozen initial rational coefficients at the current normalized input u_{in} . The input projection and the downstream cotangent remain at their current values. Products between the live and initial Jacobian kernels are evaluated directly from equation 7.

For a nonzero Jacobian with singular values $\{\sigma_j\}_{j=1}^w$,

$$\text{pr}_A(J) = \frac{(\sum_j \sigma_j^2)^2}{w \sum_j \sigma_j^4}, \quad \text{pr}_B(h) = \frac{\|h\|_2^4}{w \|h\|_4^4}. \quad (42)$$

We take $\text{pr}_A(0) = \text{pr}_B(0) = 0$. If either role has zero total sensitivity, we set both participation values to one. If either kernel-alignment denominator vanishes, we set both congruence values to one. In either case equation 18 gives $e_i^r = 0$ whenever the corresponding response comparison is undefined.

Lemma 1 (Response-summary bounds). *For every nonzero response, $\text{pr}_A(J), \text{pr}_B(h) \in [1/w, 1]$. Whenever the denominator in equation 17 is positive, $\kappa_i^r \in [0, 1]$.*

Proof. For nonnegative v_j , Cauchy–Schwarz gives $(\sum_j v_j)^2 \leq w \sum_j v_j^2$, while $(\sum_j v_j)^2 \geq \sum_j v_j^2$. Apply this first to $v_j = \sigma_j^2$ and then to $v_j = h_j^2$. For congruence, concatenate the weighted kernels as direct-sum matrices $\bigoplus_n \sqrt{\alpha_{in}^r} \Gamma_{in}^r$ and $\bigoplus_n \sqrt{\alpha_{in}^r} \Gamma_{in}^{r,0}$. Frobenius Cauchy–Schwarz gives the upper bound. The numerator is nonnegative because $\langle J J^\top, J^0 J^{0\top} \rangle = \|J^\top J^0\|_F^2$ and $\langle h h^\top, h^0 h^{0\top} \rangle = (h^\top h^0)^2$. $\square$

Proposition 2 (Norm preservation). *Assume $e \in [0, 1]$, $p \neq 0$, and $a^\perp \neq 0$. The candidate in equation 20 satisfies $\|\tilde{q}\|_F = \|p\|_F$, and the selected direction in equation 21 satisfies $\|q\|_F = \|p\|_F$.*

Proof. By construction, $\langle p, a^\perp \rangle = 0$. Since $e \in [0, 1]$,

$$\|\tilde{q}\|_F^2 = (1 - e) \|p\|_F^2 + e \|p\|_F^2 = \|p\|_F^2. \quad (43)$$

The fallback branch in equation 21 is $q = p$. $\square$

C FUNCTIONAL SCORES AND TRACE MATCHING

Let $A_i(\alpha) = A_i + \alpha q_i^A$ and $B_i(\alpha) = B_i + \alpha q_i^B$. At points where the rational response is differentiable, differentiating the group contribution $B_i(\alpha) \Phi_i(A_i(\alpha)x)$ at $\alpha = 0$ gives

$$\left. \frac{d}{d\alpha} B_i(\alpha) \Phi_i(A_i(\alpha)x) \right|_{\alpha=0} = q_i^B h_i + B_i J_i q_i^A x, \quad (44)$$

$$\left. \frac{d\mathcal{L}}{d\alpha} \right|_{\alpha=0} = \langle \xi, q_i^B h_i + B_i J_i q_i^A x \rangle. \quad (45)$$

At a zero coordinate of the normalized response, the right-hand sides define the automatic-differentiation-selected linearization from Appendix A, rather than a classical two-sided derivative. Moving the linear maps across the inner product yields equation 23. Thus one row of S records how the same loss probe responds to every layer–group direction; its rowwise outer product couples coordinates that affect that probe together.

The captured cotangent is put on this per-position scale explicitly. If a microbatch contains R activation positions, $M = 4$ microbatches are accumulated, $\xi_{k\ell}$ is the cotangent captured from the mean-reduced backward pass, and χ_{clip} is the realized clipping factor, then

$$\xi_{k\ell} = RM \chi_{\text{clip}} \bar{\xi}_{k\ell}. \quad (46)$$

The indices used to sample x , z , h , and $\bar{\xi}$ are identical within each probe row.

Lemma 2 (Trace-matched intermediate factor). *Let $\nu^2 = \|S_{\text{ref}}\|_F^2 / K$ and $s_t \neq 0$. For $S_t = \mathbf{1}_K (\nu s_t / \|s_t\|_2)^\top$,*

$$\frac{1}{K} \|S_t\|_F^2 = \nu^2. \quad (47)$$

Proof. All K rows have squared norm $\nu^2 \|s_t\|_2^2 / \|s_t\|_2^2 = \nu^2$. $\square$

Consequently, the repeated surrogate has the same trace $\text{tr}(S_t^\top S_t / K) = \nu^2$ as the latest functional score factor. It has rank at most one (rank one when $\nu > 0$) and is aligned with the current coordinate-score vector; the next functional refresh replaces that directional approximation with probe-level cross-coordinate structure.

D ROBUST FREQUENT-DIRECTIONS MIDPOINT COMPLETION

Proposition 3 (Positive semidefiniteness and spectral error). *Let $K > 0$, $0 \leq \beta_2 \leq 1$, $\tau_{t-1} \geq 0$, and $\mathcal{A} = U \text{Diag}(\sigma) V^\top$. If $\text{rank}(\mathcal{A}) > r$, set $\Delta = \sigma_{r+1}^2$ and define*

$$\mathcal{B} = \text{Diag}\left(\sqrt{(\sigma_1^2 - \Delta)_+}, \dots, \sqrt{(\sigma_r^2 - \Delta)_+}\right) V_{1:r}^\top. \quad (48)$$

If $\text{rank}(\mathcal{A}) \leq r$, set $\Delta = 0$ and choose any factor with at most r rows satisfying $\mathcal{B}^\top \mathcal{B} = \mathcal{A}^\top \mathcal{A}$. Let

$$G = \mathcal{A}^\top \mathcal{A} / K + \beta_2 \tau_{t-1} I \quad (49)$$

be the pre-compression metric, and form $\hat{G} = \mathcal{B}^\top \mathcal{B} / K + (\beta_2 \tau_{t-1} + \Delta / (2K)) I$. Then

$$\hat{G} \succeq 0, \quad \|\hat{G} - G\|_2 \leq \frac{\Delta}{2K}. \quad (50)$$

Proof. When $\text{rank}(\mathcal{A}) \leq r$, the two metrics are identical and the claim follows with $\Delta = 0$. Otherwise, the retained eigenvalues after shrinkage are nonnegative, and $\tau_t = \beta_2 \tau_{t-1} + \Delta / (2K) \geq 0$, proving positive semidefiniteness. On a retained right-singular direction, the represented eigenvalue differs from the live eigenvalue by $-\Delta / (2K)$. On a discarded direction with live eigenvalue $\lambda \in [0, \Delta]$, the error is $(\Delta / 2 - \lambda) / K$. On a null direction it is $\Delta / (2K)$. The largest absolute eigenvalue error is therefore at most $\Delta / (2K)$. $\square$

The scalar $\Delta / (2K)$ is the midpoint of the unrepresented interval $[0, \Delta / K]$ and minimizes the worst absolute scalar-completion error over that interval. The one-step result also yields a discounted history bound. Let

$$G_t^* = \beta_2 G_{t-1}^* + (1 - \beta_2) S_t^\top S_t / K, \quad G_1^* = S_1^\top S_1 / K \quad (51)$$

be the uncompressed exponentially weighted metric. If $E_t = \hat{G}_t - G_t^*$, then

$$\|E_t\|_2 \leq \sum_{j=2}^t \beta_2^{t-j} \frac{\Delta_j}{2K}. \quad (52)$$

Indeed, the compression error at step t is a perturbation D_t with $\|D_t\|_2 \leq \Delta_t / (2K)$, so $E_t = \beta_2 E_{t-1} + D_t$ and $E_1 = 0$.

E EQUAL-BUDGET TRANSACTION

For $\eta > 0$, $\omega_i > 0$, and $\delta_t \geq 0$, let $W = \text{Diag}(\omega)$ and set $y = W^{1/2} c$. Substituting $c = W^{-1/2} y$ into equations 31 and 32, then dividing by η^2 , gives the equivalent problem

$$\min_{\|y\|_2^2 = \mathcal{E}_0} \frac{1}{2} y^\top M_t y - b_t^\top y, \quad (53)$$

with M_t and b_t from equation 33. Its Lagrangian

$$\mathcal{J}(y, \lambda) = \frac{1}{2} y^\top M_t y - b_t^\top y + \frac{\lambda}{2} (\|y\|_2^2 - \mathcal{E}_0) \quad (54)$$

has stationarity equation $(M_t + \lambda I)y = b_t$.

The transaction requires $b_t \neq 0$. Its fully reorthogonalized Lanczos recurrence starts from $b_t / \|b_t\|_2$ and constructs an orthonormal basis Q_k , with $k \leq 32$. Without breakdown, this basis spans $\mathcal{K}_k(M_t, b_t) = \text{span}\{b_t, M_t b_t, \dots, M_t^{k-1} b_t\}$. If a numerical breakdown threshold is reached early, a deterministic cosine vector is reorthogonalized against the existing basis to start the next block, and the coupling between the two blocks is set to zero. The recurrence is accepted only after its final orthogonality check. Let $\bar{T}_k$ denote the symmetric block-tridiagonal matrix accumulated by this recurrence; without a threshold restart, $\bar{T}_k = Q_k^\top M_t Q_k$. If $\bar{T}_k = U \text{Diag}(\bar{\theta}) U^\top$, define the numerical positive-semidefinite surrogate $T_k^+ = U \text{Diag}(\theta) U^\top$ with $\theta_j = \max\{\bar{\theta}_j, 0\}$. The solver uses

$$(T_k^+ + \lambda I)z = Q_k^\top b_t, \quad \|z\|_2^2 = \mathcal{E}_0. \quad (55)$$

With $\gamma = U^\top Q_k^\top b_t$, the norm constraint becomes

$$\psi(\lambda) = \sum_j \frac{\gamma_j^2}{(\theta_j + \lambda)^2} = \mathcal{E}_0. \quad (56)$$

On the regular branch $\lambda > 0$, ψ is continuous and strictly decreasing because $\gamma \neq 0$ and $\theta_j \geq 0$. Let

$$\lambda_{\text{lo}} = \epsilon_{\text{mach}} C, \quad \lambda_{\text{hi}} = \frac{\|b_t\|_2}{\sqrt{\mathcal{E}_0}} + \lambda_{\text{max}}(T_k^+) + \lambda_{\text{lo}}. \quad (57)$$

Because $\|\gamma\|_2 = \|b_t\|_2$ and every $\theta_j \geq 0$, $\psi(\lambda_{\text{hi}}) \leq \|b_t\|_2^2 / \lambda_{\text{hi}}^2 \leq \mathcal{E}_0$. When $\psi(\lambda_{\text{lo}}) \geq \mathcal{E}_0$, a unique root in the positive bracket is approximated by 64 bisection steps. Otherwise no multiplier in the implemented interval $[\lambda_{\text{lo}}, \infty)$ attains the budget. TILLER does not use negative-multiplier stationary points, for which the shifted system can approach singularity; it retains the parent transaction in this case.

The projected coefficient vector is evaluated in double precision using the unprojected low-rank-plus-diagonal objective. A finite proposal is retained when

$$\frac{|c^\top Wc - \mathcal{E}_0|}{\max\{\mathcal{E}_0, 1\}} \leq 10^{-8}, \quad F_t(c) < F_t(\mathbf{1}), \quad (58)$$

and both its and the parent’s aggregate current-gradient and Nesterov contractions are positive for both matrix roles in each layer. The residual check makes acceptance independent of a uniform error guarantee for the fixed bisection count.

F ONE TILLER UPDATE

For completeness, the five-step polar map starts from the input matrix normalized by its Frobenius norm (lower bounded by 10^{-7}), transposing tall matrices before the iteration. Each step is

$$X_{j+1} = aX_j + b(X_j X_j^\top)X_j + c(X_j X_j^\top)^2 X_j, \quad (a, b, c) = (3.4445, -4.775, 2.0315), \quad (59)$$

after which the original orientation is restored.

The state at step t comprises Nesterov momentum, factorized second moments, the most recent response summaries, the score factor $\mathcal{B}_{t-1}$, its isotropic tail τ_{t-1} , and the score–decay cross term. One update proceeds as follows. The factorized moments use float32 accumulators and $\delta_{\text{fp}} = \text{finfo}(\text{float32}).\text{tiny}$; when the row-moment total is zero, the estimate in equation 12 is $V_t = 0$.

1. Update momentum and construct the five-step Newton–Schulz Muon parent directions for the paired GRAIN matrices and the attention matrices.
2. Update the factorized second moments. On a functional refresh, compute participation and congruence; between refreshes, reuse their latest values. Form the response-gated tangent chords.
3. At step 1 and every eight steps thereafter, evaluate the 32 rows in equation 23 and the corresponding decay action. At other steps, form the trace-matched rows from the current batch-gradient scores.
4. Update the rank-64 Robust Frequent-Directions factor, isotropic tail, and score–decay cross term.
5. Construct the at-most 32-dimensional Lanczos basis, attempt the positive regular branch of equation 56, and evaluate the proposed transaction.
6. Scale both GRAIN matrix directions in coordinate (ℓ, g) by $c_{\ell g}$, then apply the scheduled parameter update and decoupled decay. Apply the response-informed chord to the attention matrices.
7. Update the rational coefficients and remaining parameter classes with AdamW.

The attention polar map groups 16 heads in blocks of four. GRAIN uses group width $w = 256$. Undefined response comparisons set $e = 0$, degenerate tangents select the Muon parent, and unbracketed positive-multiplier branches or rejected proposals retain $c = 1$.

G COMPUTATION AND PARTITIONED EVALUATION

Let device j hold a set of coordinate columns $\mathcal{A}_t^{(j)}$ and the corresponding segments of s_t , ω , δ_t , and r_t . The identities

$$\mathcal{A}_t \mathcal{A}_t^\top = \sum_j \mathcal{A}_t^{(j)} \mathcal{A}_t^{(j)\top}, \quad \langle u, v \rangle = \sum_j \langle u^{(j)}, v^{(j)} \rangle \quad (60)$$

express the row Gram matrix, Krylov inner products, norms, and transaction budget as sums of local contributions. Each coordinate segment can apply its coefficient directly after the collective scalar and row-space quantities have been combined.

With $C = LG$, probe count K , ledger rank r , Krylov dimension k , and $m = r + K$, the ledger stores $O(rC)$ values. One update, including row-space diagnostics, costs

$$O((m^2 + km + k^2)C + m^3 + k^3). \quad (61)$$

The m^2C and m^3 terms form and diagonalize the temporal row Gram matrix; the kmC , k^2C , and k^3 terms construct, reorthogonalize, and solve in the projected transaction space. The factorized moments add $O(LH_r + LGd)$ state. In TILLER, $K = k = 32$ and $r = 64$, so the fixed-rank terms scale linearly with the number of layer-group coordinates. The activation computation scales as $O(LNH_r)$, while persistent optimizer state is independent of the number of activation positions.